

# A Short Note on Scaled Dot-Product Attention

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## Abstract

Attention lets every token gather information from every other token in a sequence.

This note restates the scaled dot-product attention used by the Transformer.

## 1 Method

Queries  $Q$ , keys  $K$  and values  $V$  are linear projections of the token embeddings.

The output is  $\text{softmax}(Q K^T / \sqrt{d_k}) V$ , where  $d_k$  is the key dimension.

Scaling by the square root of the key dimension keeps the softmax from saturating.

Without the scaling, dot products grow with  $d_k$  and gradients through the softmax vanish.

## **2 Multi-Head Attention**

Running  $h$  attention heads in parallel lets the model attend to different subspaces.

The heads are concatenated and projected back to the model dimension.

## **3 Conclusion**

Scaled dot-product attention is simple, parallel, and the core of the Transformer.